

Output Feedback Adaptive Optimal Control of Affine Nonlinear Systems with a Linear Measurement Model

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Outline

Motivation

Problem Formulation

State Estimator Design

Control Design

Simulation

Summary

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- ▶ Problem: In many robotic systems, sensor information is not always available, which affects the performance of controllers due to incomplete state information.
- ▶ Idea: Reinforcement learning (RL) controllers are robust to modeling errors.
- ▶ Challenges: Common RL schemes in continuous time and space are
 - ▶ resource intensive
 - ▶ requires offline learning
 - ▶ stability while learning is generally not guaranteed
- ▶ Model-based reinforcement learning (MBRL) adaptive optimal control
 - ▶ Online learning with stability guarantees
 - ▶ Requires accurate estimate of the model to satisfy excitation conditions and guarantee stability

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- Nonlinear output feedback dynamical system:

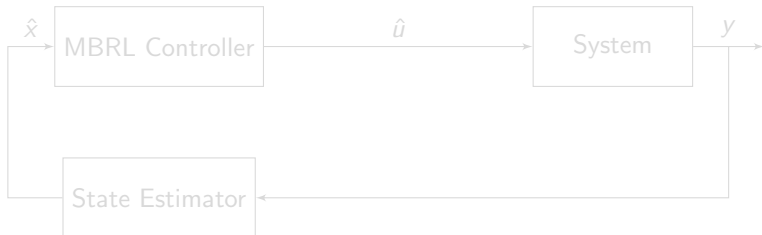
$$\dot{x} = f(x) + g(x)u, \quad y = Cx$$

- Objectives:

- Design an observer to estimate the states online, using input-output measurements
- Simultaneously synthesize and utilize a controller that minimizes the cost functional

$$\min_{u \in \mathcal{U}_{[t_0, \infty)}} \int_{t_0}^{\infty} r(x(\tau), u(\tau)) d\tau$$

while ensuring local uniform ultimate boundedness of the trajectories of the system



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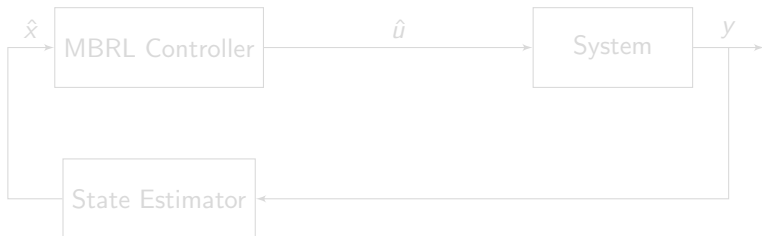
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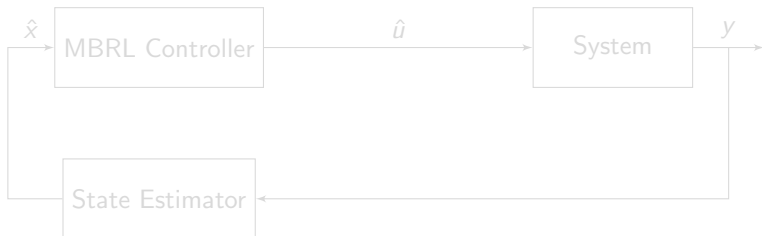
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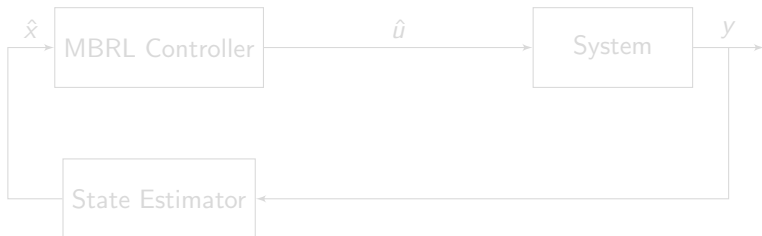
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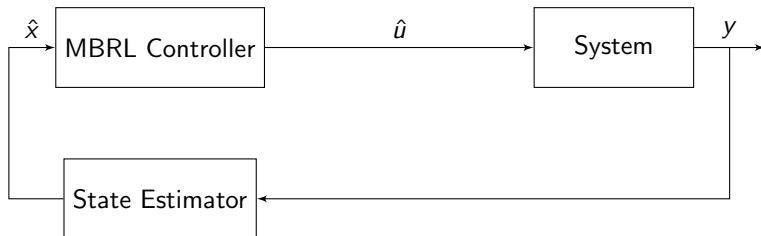
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Assumption 1: Bounded Jacobian

The functions f and g are known, their derivatives exist on a compact set $\mathcal{C} \subset \mathbb{R}^n$ containing the origin, and satisfy the element-wise bounds

$$(M_{f_1})_{i,j} \leq \frac{\partial(f(x))_i}{\partial(x)_j} \leq (M_{f_2})_{i,j},$$
$$(M_{g_1})_{i,j} \leq \left(\frac{\partial(g(x))_{i,k}}{\partial(x)_j} \right) (u)_k \leq (M_{g_2})_{i,j},$$

for all $x \in \mathcal{C}$, $|(u)_k| \leq \lambda$, $i, j = 1, \dots, n$ and $k = 1, \dots, m$, where $(\cdot)_i$ and $(\cdot)_{i,j}$ denote the element of the array $(\cdot)$ at the indices indicated by the subscript.

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State estimator

- State estimator with three correction terms:

$$\dot{\hat{x}} = A\hat{x} + \bar{f}[\hat{x} + H(y - C\hat{x})] + \bar{g}_u[\hat{x} + K(y - C\hat{x}), u] + L(y - C\hat{x})$$

where $A := M_{f_1} + M_{g_1}$, $\bar{f}(x) = -M_{f_1}x + f(x)$, and
 $\bar{g}_u(x, u) = -M_{g_1}x + \sum_{i=1}^m g_i(x)(u)_i$

- $H \in \mathbb{R}^{n \times q}$, $K \in \mathbb{R}^{n \times q}$ and $L \in \mathbb{R}^{n \times q}$ are observer gains
- Under Assumption 1, the derivatives of $\bar{f}$ and $\bar{g}$ satisfy

$$0 \leq \frac{\partial(\bar{f}(x))_i}{\partial(x)_j} \leq (M_{f_2})_{i,j} - (M_{f_1})_{i,j}, \text{ and}$$

$$0 \leq \frac{\partial(\bar{g}_u(x, u))_i}{\partial(x)_j} \leq (M_{g_2})_{i,j} - (M_{g_1})_{i,j}$$

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Multiplier Formulation and Sector Conditions

- Estimation error: $e := x - \hat{x}$
- Using differential mean value theorem (DMVT):

$$\overline{M}_{f_1}(I - HC)e \leq \phi_f(t, e) \leq \overline{M}_{f_2}(I - HC)e, \text{ and}$$

$$\overline{M}_{g_1}(I - KC)e \leq \phi_g(t, e, u) \leq \overline{M}_{g_2}(I - KC)e.$$

where $\phi_f(t, e) := \bar{f}(x) - \bar{f}[\hat{x} + H(y - C\hat{x})]$,
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State estimator analysis using Lyapunov theory

- Lyapunov function: $V_e(e) := e^T P e$
- Orbital derivative of the Lyapunov function:

$$\begin{aligned}\dot{V}_e(e) = & \begin{bmatrix} e \\ \phi_f \end{bmatrix}^T \begin{bmatrix} \left(M_{f_1} - \frac{LC}{2}\right)^T P + P \left(M_{f_1} - \frac{LC}{2}\right) & P \\ P & 0 \end{bmatrix} \begin{bmatrix} e \\ \phi_f \end{bmatrix} \\ & + \begin{bmatrix} e \\ \phi_g \end{bmatrix}^T \begin{bmatrix} \left(M_{g_1} - \frac{LC}{2}\right)^T P + P \left(M_{g_1} - \frac{LC}{2}\right) & P \\ P & 0 \end{bmatrix} \begin{bmatrix} e \\ \phi_g \end{bmatrix}.\end{aligned}$$

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Stability analysis of observer error system

Theorem 1

Given a system satisfying Assumption 1, provided the control input remains bounded and the system trajectories remain within the compact set $\mathcal{C}$, if there exists a symmetric positive definite matrix P , and observer gains L , H , K that satisfy the matrix inequality

$$\begin{bmatrix} \left((A - LC)^T P + P(A - LC) \right) + 2\alpha P & P - J_{fh21}^T & P - J_{gk21}^T \\ P - J_{fh21} & -(J_f)_{22} & 0 \\ P - J_{gk21} & 0 & -(J_g)_{22} \end{bmatrix} \leq 0,$$

where

$$J_f := \begin{bmatrix} 0 & -\frac{M_{f2}^T - M_{f1}^T}{2} \\ -\frac{M_{f2} - M_{f1}}{2} & I \end{bmatrix}, J_g := \begin{bmatrix} 0 & -\frac{M_{g2}^T - M_{g1}^T}{2} \\ -\frac{M_{g2} - M_{g1}}{2} & I \end{bmatrix},$$

$J_{fh21} := (J_f)_{21} (I - HC)$ and $J_{gk21} := (J_g)_{21} (I - KC)$, then the observer error system is locally uniformly asymptotically stable.

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Optimal Control Problem

- Cost functional:

$$J(x, u(\cdot)) := \int_0^\infty Q(x) + U(u(\tau)) d\tau$$

- State Penalty: $Q : \mathbb{R}^n \rightarrow \mathbb{R}$ is a continuous, positive definite function.
- To enforce the saturation constraint on the control:

$$U(u) := 2 \int_0^u (\lambda \tanh^{-1}(v/\lambda))^T R dv,$$

- Objective: minimize cost functional subject to system constraints.

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Optimal Control Problem

- Assuming the optimal controller exists, the **optimal value function** is defined as

$$V^*(x) := \min_{u(\cdot) \in \mathcal{U}_{[t, \infty)}} \int_t^\infty Q(x) + U(u(\tau)) d\tau.$$

- Assuming that the optimal value function is continuously differentiable, it can be obtained by solving the corresponding Hamilton-Jacobi Bellman (HJB) equation

$$\min_{u \in \mathbb{R}^m} \left(\nabla_x V^*(x) (f(x) + g(x)u) + Q(x) + U(u) \right) = 0,$$

- Optimal Controller:

$$u^*(x) := -\lambda \tanh(D^*)$$

where

$$D^* = \frac{1}{2\lambda} R^{-1} g(x)^T \nabla_x V^*(x) \in \mathbb{R}^m$$

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Value Function Approximation

- ▶ Problem: Solving the HJB equation for nonlinear systems is generally infeasible
- ▶ Solution: Use approximation property of neural networks to approximate the value function

$$V^*(x) = W^T \sigma(x) + \epsilon(x)$$

- ▶ Since the ideal weights, W , are unknown, we use estimates $\hat{W}_c$ called critic weights
- ▶ Approximate Value Function: $\hat{V}(\hat{x}, \hat{W}_c) := \hat{W}_c^T \sigma(\hat{x})$
- ▶ Approximate control policy: $\hat{u}(\hat{x}, \hat{W}_c) := -\lambda \tanh(\hat{D})$ where

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$$V^*(x) = W^T \sigma(x) + \epsilon(x)$$

- ▶ Since the ideal weights, W , are unknown, we use estimates $\hat{W}_c$ called critic weights
- ▶ Approximate Value Function: $\hat{V}(\hat{x}, \hat{W}_c) := \hat{W}_c^T \sigma(\hat{x})$
- ▶ Approximate control policy: $\hat{u}(\hat{x}, \hat{W}_c) := -\lambda \tanh(\hat{D})$ where

$$\hat{D} = \frac{1}{2\lambda} R^{-1} g(\hat{x})^T \nabla_{\hat{x}} \sigma(\hat{x})^T \hat{W}_c$$

Bellman Error and Update Laws

- The residual term referred to as the Bellman error (BE):

$$\hat{\delta}(\hat{x}, \hat{W}_c) = \nabla_x \hat{V}(\hat{x}, \hat{W}_c) \left(f(\hat{x}) + g(\hat{x})\hat{u}(\hat{x}, \hat{W}_c) \right) + U(\hat{u}) + Q(\hat{x}).$$

- BE extrapolation ensures excitation for simulation, ensuring stability and convergence of online RL.
- Simulation of experience in a model-based RL scheme by using the system model to extrapolate the approximate BE to unexplored areas of the state space.
- Update laws for Critic weights and Learning gains:

$$\dot{\hat{W}}_c = -\frac{k_c}{N} \Gamma \sum_{i=1}^N \frac{\omega_i}{\rho_i} \hat{\delta}_i, \quad \dot{\Gamma} = \beta \Gamma - \frac{k_c}{N} \Gamma \sum_{i=1}^N \frac{\omega_i \omega_i^T}{\rho_i^2} \Gamma$$

where $\omega_i := \nabla_{x_i} \sigma(x_i) \left(f(x_i) + g(x_i)\hat{u}(x_i, \hat{W}_c) \right)$,
 $\rho_i(t) := 1 + \gamma \omega_i^T(t) \omega_i(t)$.

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Stability Analysis

Assumption 2

There exists a constant $\underline{c}_1$ such that the finite set of trajectories $\{x_i : \mathbb{R}_{\geq t_0} \mid i = 1, \dots, N\}$ satisfies

$$0 < \underline{c}_1 \leq \inf_{t \in \mathbb{R}_{\geq T}} \lambda_{\min} \left(\frac{1}{N} \sum_{i=1}^N \frac{\omega_i(t) \omega_i^T(t)}{\rho_i^2(t)} \right). \quad (1)$$

Theorem 2

Provided assumptions 1 and 2 holds, there exists a symmetric positive definite matrix P , and observer gains L, H, K that satisfy the matrix inequality in Theorem 1, the control gains are selected large enough based to satisfy the subsequently developed sufficient conditions, and the weights $\hat{W}_c$ and Γ are updated according to the designed update laws, then the trajectories of the closed-loop system are locally uniformly ultimately bounded under the developed controller.

Outline

Motivation

Problem Formulation

State Estimator Design

Control Design

Simulation

Summary

Simulation

- ▶ A simulation study was performed on a control affine nonlinear system with states $x = [(x)_1; (x)_2]$ and of the form
- ▶
$$f(x) = \begin{bmatrix} -(x)_1 + (x)_2 \\ -(x)_1 - \frac{1}{2}(x)_2 \left(1 - (\cos(2(x)_1) + 2)^2\right) \end{bmatrix},$$
$$g(x) = \begin{bmatrix} 0 \\ \cos(2(x)_1) + 2 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 1 \end{bmatrix}$$
- ▶ The LMI was solved using Sedumi in YALMIP on Matlab with the learning rate, $\alpha = 2$, control bound, $\lambda = 3$
- ▶ The basis for value function approximation is selected as $\sigma(\hat{x}) = [(\hat{x})_1^2; (\hat{x})_1(\hat{x})_2; (\hat{x})_2^2]$.
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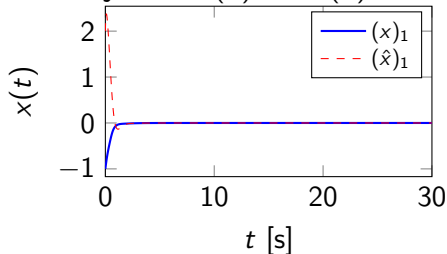
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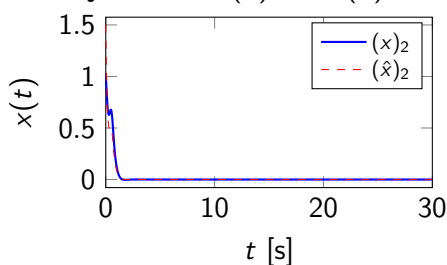
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Simulation Results

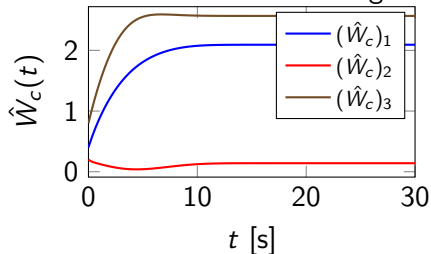
Trajectories $(x)_1$ and $(\hat{x})_1$



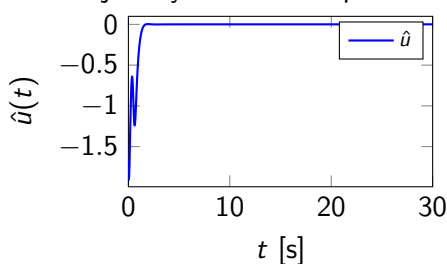
Trajectories of $(x)_2$ and $(\hat{x})_2$



Estimated value function weights



Trajectory of control input



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Thank You!